F5 BIG-IP DNS

DNS Management Portal

Requirements Specification

Requirements gathering · Version 1.0 · 05-Nov-2026

Text highlighted in yellow is an open point that still needs a decision or an input. All open points are also collected in section 15.

Document control	
Document	DNS Management Portal – Requirements Specification
Version / date	1.0 · 05-Nov-2026
Phase	Requirements gathering: for review and confirmation by SBI and Starlight Data Solutions. The open points in section 15 are still to be closed.
Based on	RFP Section J (J1–J5), portal specification extract Solution Proposal v2.2 (04-Nov-2026), the agreed baseline for all updates since v2.0 The Bank's answers of 04-Nov-2026 Requirement discussions and decisions of 03–05 Nov 2026

1. Purpose and scope

SBI is deploying F5 BIG-IP DNS (GTM) on VELOS CX410 at the DC, DR and AZ sites. The DNS Management Portal is the single governed front end through which Bank users view and change DNS records on these devices, as required by RFP Section J. DNS resolution stays on F5 and never depends on the portal.

This document records the requirements gathered so far and the points that are still open. It will be updated as the open points are closed.

1.1 Scope

In scope (this phase)	Later, or not decided yet
• Sign-in with portal accounts and the Bank directory (LDAP) • User onboarding and access requests • Access levels Read, Read + Write, Checker, Admin, scoped by F5 partition • DNS records: one-to-one, one-to-many and PTR, for IPv4 (A records) and IPv6 (AAAA records) • Maker–checker approval and direct Checker/Admin changes • Scheduling of updates to existing records • Versioning, version comparison and rollback • Result per F5 site (DC / DR / AZ) with the F5 error shown on failure • Dashboards and reports (RFP J1.11); in-app notifications	• Other record types (CNAME, MX, TXT, NS, SRV) • Wide IPs, pools, monitors, topology (object scope) • Fetch and sync all records from F5 (J1.5) • ITSM integration (on hold at the Bank) • MFA (removed from scope; deviation needed)

In scope (this phase)	Later, or not decided yet
• Audit trail, log files and SIEM forwarding	

2. Definitions

Term	Meaning
Partition	An F5 BIG-IP partition. Every DNS record belongs to exactly one partition. Partitions decide what a Read + Write user may change.
Department	The team a user works in (e.g. Network Operations). Information only: it grants no access. Entered by the user, because the Bank directory doesn't provide it.
User ID	The numeric employee ID used to sign in (e.g. 140122).
A / AAAA record	A DNS record that points a name to an IPv4 address (A) or an IPv6 address (AAAA). The portal picks the type from the address entered.
Bank directory	The Bank LDAP directory. It confirms who a user is (ID, name, email, password). The portal never stores Bank passwords.
Local user	A portal account created by the Admin, with a password kept (hashed) by the portal.
Admin	The one configuration-defined administrator account.
Maker / Checker	A maker raises a change; a Checker (or the Admin) approves or rejects it.
Direct change	A change by a Checker or the Admin. It is validated and applied without a second approval.
Version	A saved, unchangeable snapshot of a record after each successful change: v1.0, v1.1, v1.2 …
Rollback	Restoring the configuration of an earlier version as a new version (history is never rewritten).
Held	A scheduled rollback that was not run because the record changed in the meantime; it waits for the Admin.
Active / Standby site	Only the Active site (DC normally, DR after failover) runs changes and scheduled jobs.

3. Users, roles and access levels

3.1 Parties

Party	Role in the project
SBI (the Bank)	Owner and users of the portal. Provides the LDAP directory, ITSM, SIEM, Vault/KMS and the private-cloud VMs.
F5	OEM and bidder. Provides BIG-IP DNS, the API documentation and the UAT device.
Starlight Data Solutions	Builds, delivers and supports the portal.
LB team	Maintains the Bank's load balancers: portal VIPs, GSLB, health checks, TLS and WAF.
Network Integrator	The Bank's network partner; receives the knowledge transfer.

3.2 Access levels

The Admin gives each user exactly one access level. Permissions are checked by the server on every request, not only hidden in the screens; attempts outside a user’s access are refused and recorded as denied.

Capability	Read	Read + Write	Checker	Admin
View, search records and version history	All partitions	Own partitions	All partitions	All partitions
Create, modify, delete records	No	Own partitions, needs approval	All, direct	All, direct
Schedule an update to an existing record	No	Yes, approved first	Yes	Yes
Edit or cancel own pending / scheduled change	No	Yes, until it runs	Yes	Yes
Approve or reject Write requests	No	No	Yes, not own	Yes, not own
Compare versions and roll back	No	No	No	Yes
Users, access requests, partitions, audit trail	No	No	No	Yes

4. Sign-in, accounts and sessions

ID	Requirement
AUTH-01	One Admin account, defined in protected configuration (preferably injected from the Bank Vault/KMS) and stored only as a salted hash. It cannot be created, edited or deleted in the screens.
AUTH-02	Sign-in checks portal accounts first (Admin, local users, known directory users). The Bank directory is asked only for IDs the portal does not know.
AUTH-03	Users sign in with their user ID, the numeric employee ID (e.g. 140122). **Open:** confirm that the LDAP user ID (uid) is the employee ID.
AUTH-04	The portal never keeps a Bank user's password. Each time a directory user signs in, the portal asks the Bank directory to check the password. If the directory is unavailable, directory users cannot sign in; the Admin and local accounts still can (as in proposal v2.2).
AUTH-05	Local users get a one-time temporary password from the Admin and must set their own password at first sign-in.
AUTH-06	Password rules for local accounts: at least 12 characters with upper case, lower case, digit and special character; must not contain the user ID; expires after 90 days. **Open:** proposed values; to be replaced by the Bank IS password policy (length, history, expiry).
AUTH-07	Five failed sign-ins lock the account for 15 minutes; the Admin can unlock it. **Open:** confirm the values with the Bank IS policy.
AUTH-08	Server-side sessions: the session cookie is HttpOnly, SameSite=Strict and Secure over HTTPS; a new session ID at sign-in; 30 minutes idle timeout; sign-out ends the session. Permissions are re-read on every request, so disabling a user or changing access applies at once.
AUTH-09	Further session controls: a maximum session length, a limit on simultaneous sign-ins, an Admin view to see and end active sessions, and ending other sessions when a password changes. **Open:** which of these are needed (Bank IS policy).
AUTH-10	Sign-in attempts are rate-limited at the web tier.
AUTH-11	Multi-factor authentication (MFA). **Open:** removed from scope on 03-Nov-2026, but RFP J1.2 asks for MFA for all users. A deviation must be agreed with SBI (via F5).

5. User onboarding and access requests

ID	Requirement
ONB-01	The first successful directory sign-in of an unknown user creates a profile with no access ("Pending access"). The user sees no DNS data.

ID	Requirement
ONB-02	The pending user raises an access request in the portal: department, access needed (Read only / Read + Write / Checker), partitions (for Read + Write) and a reason. The request can be edited while it is open.
ONB-03	The Bank directory provides only ID, name and email. The department is entered by the user, can be edited by the Admin and is kept by the portal.
ONB-04	The Admin sees requests under Pending access (with a count on the menu and the dashboard). The grant form starts from what was asked; the Admin grants it as asked, adjusts it, or declines it with a reason the user sees. A declined user can ask again. The audit trail records both what was asked and what was granted.
ONB-05	The Admin can look up any ID in the Bank directory and grant access before the user ever signs in.
ONB-06	The Admin can create local users, change access, disable and enable users, unlock accounts and reset local passwords. Disabling a waiting user closes their request.
ONB-07	Notifications are in the portal only (no e-mail): the Admin sees new access requests as a count on the menu and the dashboard; the user sees the decision (access, or the reason it was declined) at the next sign-in. Checkers see requests waiting for them the same way.
ONB-08	Suggest partitions from the department: when the Admin gives access, the partitions that other people in the same department have are ticked in advance. The Admin can change them before saving. **Open:** do we want this?

6. Partitions and departments

ID	Requirement
PAR-01	Each partition is an F5 BIG-IP partition, and every DNS record belongs to exactly one partition. **Open:** confirm with F5 how partitions apply to the DNS objects the portal manages (e.g. zone records and PTR records versus Wide IPs).
PAR-02	The Admin creates, renames and deletes partitions in the portal. Names follow the F5 rule: start with a letter; letters, digits, underscore, dash or dot; no spaces; at most 64 characters; unique ignoring case. A partition that owns records or change history cannot be deleted.
PAR-03	Read + Write users are given one or more partitions and only see and change those. Read, Checker and Admin cover all partitions.
PAR-04	Department is information only and grants no access.
PAR-05	Read the list of partitions from F5, in addition to the Admin maintaining it. **Open:** needs the F5 API.
PAR-06	The Admin moves a record to another partition (record ownership, as in the permission matrix of proposal v2.2).

7. DNS records

ID	Requirement
DNS-01	Mapping types: one-to-one (a name to one address), one-to-many (a name to 2 to 32 addresses) and PTR (an address back to one name; the reverse name is generated).
DNS-02	Addresses may be IPv4 or IPv6. The portal picks the record type from the address: A for IPv4, AAAA for IPv6; a name can have both. PTR names are generated under in-addr.arpa (IPv4) or ip6.arpa (IPv6). IPv6 addresses are validated and stored in one standard form.
DNS-03	Validation before any change: fully qualified domain name (labels 1–63 characters of letters, digits and hyphens; at most 253 characters; last label not only digits); valid IPv4 or IPv6 address (not 0.0.0.0 or 255.255.255.255); no repeated address; TTL from 60 to 86,400 seconds (default 3,600). **Open:** the Bank's DNS form fields and rules (to be asked together with the F5 API documentation).
DNS-04	One active record per name (A / AAAA) or per address (PTR); one open request per record at a time; the name (or IP for PTR) of an existing record cannot be changed (delete and create a new one).

ID	Requirement
DNS-05	Search and list records by name or IP, partition and mapping type. Deleted records are listed separately and keep their full history.
DNS-06	Further record types (CNAME, MX, TXT, NS, SRV) and CNAME loop prevention (RFP J1 validation). **Open:** which of these the Bank needs, and when.
DNS-07	Wide IPs, pools, monitors and topology records. **Open:** object scope: RFP J1.1 says "all front-end DNS management tasks"; current estimate is DNS records only.
DNS-08	Fetch and sync all records from F5 and reconcile differences (RFP J1.5). **Open:** needs the F5 API.
DNS-09	Volumes: the compliance sheet (E1) states up to 500 zones × 1 lakh records. **Open:** actual SBI zone and record counts (sizing).

8. Change workflow

ID	Requirement
CHG-01	A Read + Write user's create, modify or delete becomes a request for all Checkers and the Admin. One approval applies it. A rejection needs a comment for the maker. Nobody approves their own request (as in proposal v2.2). Multi-level approval per partition can be added if the Bank asks for it.
CHG-02	Checker and Admin changes are validated and applied directly, without a second approval, and recorded as direct changes (as in proposal v2.2). RFP J1.8 asks that the workflow can apply maker–checker; it applies to every Read + Write user.
CHG-03	The maker can edit or cancel their own request until it runs.
CHG-04	When a change runs, the portal checks again that the record is still at the version the change was prepared against. If not, the change fails safely and is not applied ("raise a new request").
CHG-05	Each change is applied at DC, DR and AZ. The F5 result is stored per site. On failure the request shows, for each site, the error code and message exactly as returned by F5, to the maker, the Checkers and the Admin. A failed change creates no version. **Open:** F5 API documentation and UAT device.
CHG-06	Execution state is saved before and after each F5 call, so a change can never be applied twice after a retry or site failover, and in-flight changes are reconciled first.
CHG-07	ITSM: create the change ticket, update its status, raise an incident on failure. **Open:** ITSM is on hold at the Bank: product, API and depth; and whether an ITSM approval must also gate execution.

8.1 Scheduled changes

ID	Requirement
SCH-01	Only updates to an existing record (modify) and Admin rollbacks can run at a chosen date and time. Creating or deleting a record is never scheduled: it runs straight away, or as soon as it is approved (decision of 05-Nov-2026).
SCH-02	The time must be at least 1 minute and at most 90 days ahead. It is shown in the user's time zone and stored in UTC.
SCH-03	A Read + Write user's scheduled update is approved first and then waits. If its time passes before anyone approves it, it cannot be approved until the maker chooses a new time. **Open:** confirm (the alternative is to run it at once on approval).
SCH-04	Editing an approved scheduled update sends it back for re-approval. The maker can cancel it until it runs.
SCH-05	Only the Active site runs scheduled jobs. Jobs are stored in the database, so they survive a failover, and a job can never run twice.
SCH-06	A scheduled update does not lock the record. If the record changes in the meantime, the scheduled change fails safely (a rollback is held for the Admin). **Open:** confirm (the alternative is to block all changes to a record while something is scheduled on it).

ID	Requirement
SCH-07	A scheduled change does not run if its maker has been disabled, or if the maker of a direct change has lost Checker access.
SCH-08	The maker can cancel an approved scheduled update until it runs. **Open:** should the Checker who approved it also be able to cancel it?

9. Versions, comparison and rollback

ID	Requirement
VER-01	A record starts at v1.0 when first created; each successful change adds 0.1. Each version is a full, unchangeable snapshot with who, when, reason, ITSM reference and the request. A delete creates a final "deleted" version.
VER-02	Everyone who can see a record sees its version history. The Admin compares any two versions field by field (added and removed addresses marked).
VER-03	The Admin rolls a record back to any earlier version, now or at a scheduled time. A rollback creates a new version (rolling 1.3 back to 1.1 creates 1.4) and also restores deleted records. It is refused while a request is waiting for approval on the record, if the configuration would not change, or if a newer record already uses the name.
VER-04	A scheduled rollback whose record changed in the meantime is held for the Admin to run or cancel.
VER-05	Before a rollback, compare the portal's latest version with the live F5 configuration and show any change made directly on F5. **Open:** needs the F5 read API.

10. Dashboard and reports

ID	Requirement
RPT-01	Dashboard: total records; requests awaiting my approval (Checker) or my pending requests (Read + Write); changes applied, failed and scheduled in the last 7, 30 or 90 days; users waiting for access (Admin); records per partition; platform health; recent requests.
RPT-02	Reporting and analytics (RFP J1.11; "dashboards and reports" in proposal v2.2): for example changes by partition, user or date, failed changes and rollbacks, with download. **Open:** which reports the Bank needs beyond the dashboard.
RPT-03	RFP J1.11 asks for dashboards of the number of records per department. In the portal design, records belong to partitions and the department describes users, so the dashboard counts records per partition. **Open:** is records per partition enough, or should records also be counted per department (for example by the department of the person who created them)?

11. Audit trail, logs and SIEM

ID	Requirement
AUD-01	Every sign-in, lockout, user and access change, access request, partition change, request, approval, schedule, execution, rollback, hold and denied attempt is recorded. The audit table is append-only: the database refuses any change or delete.
AUD-02	Events about a user carry the user ID and name. The Admin searches the trail by ID, name, action or details.
AUD-03	Audit history is kept in the portal database for 7 years (RFP reporting horizon). **Open:** archiving and purge approach after 7 years.
AUD-04	Log files: audit, application and API access logs as one JSON object per line, each with the request ID so one action can be traced across them; rotated daily and by size and compressed. Kept 365 days (audit) and 180 days (application, access; CERT-In minimum). **Open:** confirm retention with the Bank policy.

ID	Requirement
AUD-05	Fluent Bit ships the log files to the SIEM without loss or duplicates across restarts. **Open:** SIEM product, format and transport (syslog over TLS, HEC, Kafka …).

12. Integrations

System	Used for	Inputs still needed
Bank directory (LDAP)	Sign-in of Bank users; look-up by ID	LDAPS URL, base DN, user filter, service account, returned attributes, test users
F5 BIG-IP DNS	Apply changes per site; read live configuration; sync	API documentation (list already requested), BIG-IQ or direct BIG-IP, sync-group behaviour, UAT BIG-IP VE, least-privilege API account, TLS version
ITSM	Change tickets, status, incidents	On hold at the Bank: product, API, depth
SIEM	Audit and security events	Product, log format, transport
Vault / KMS	Secrets (DB, LDAP, F5, ITSM credentials)	Bank Vault/KMS details, or a dedicated Vault

13. Non-functional requirements

ID	Requirement
NFR-SEC-01	HTTPS for all user traffic, terminated at the Bank/F5 VIP. **Open:** RFP asks for TLS 1.3+, but the F5 management API supports TLS 1.2 only (compliance J3.5 = N): deviation request to SBI.
NFR-SEC-02	CSRF protection, security headers (CSP, no framing), sign-in rate limiting, server-side permission checks, no secrets in code.
NFR-SEC-03	Secrets in the Bank Vault/KMS in UAT and production (not in configuration files).
NFR-SEC-04	Annual VAPT; non-disruptive patching; critical patches within 72 hours (J4.4).
NFR-AVL-01	DC active, DR hot standby, one portal URL through the global VIP/GSLB; two portal nodes and PostgreSQL HA per site with replication to DR; an HA/DR controller for site role, database promotion and scheduler gating (proposal v2.2).
NFR-AVL-02	RTO 1 minute, RPO 15 minutes (J4.3). **Open:** does the RTO apply to one server failing or to the whole DC lost? Ask for both values.
NFR-PRF-01	Response times and volumes for the record list, search and changes. **Open:** depends on the sizing answer (DNS-09).
NFR-OPS-01	Technology: Java 21 + Spring Boot, React (TypeScript), PostgreSQL 16, NGINX, Fluent Bit, on RHEL 9 in the Bank private cloud (as in proposal v2.2).
NFR-SUP-01	24x7 support with SLAs, documentation, training and knowledge transfer, 7 years of support (J5).

14. Decision log

Date	Decision
03-Nov-2026	Sign-in uses portal accounts first, then the Bank LDAP directory; no SSO. MFA removed from scope (deviation to be agreed).
03-Nov-2026	One Admin account from protected configuration, stored as a hash, not editable in the screens.
04-Nov-2026	Permission model of proposal v2.2: Read covers all partitions; Read + Write only assigned partitions with approval; Checker covers all partitions and changes directly; Admin = Checker plus administration.
04-Nov-2026	Admin changes and rollbacks run directly, without approval.

Date	Decision
04-Nov-2026	F5 errors are shown exactly as returned, per site; no extra plain-language hints.
04-Nov-2026	Groups are created by the Admin in the portal (no Bank list needed); a group is an F5 partition.
04-Nov-2026	The directory is LDAP (LDAPS bind). ITSM integration is on hold at the Bank.
04-Nov-2026	Users sign in with the numeric employee ID; the Admin can look up an ID in the directory and grant access before the first sign-in.
04-Nov-2026	Logs are kept as JSON files with rotation and retention and forwarded to the SIEM.
05-Nov-2026	The directory does not provide the department: the user enters it in an access request raised in the portal.
05-Nov-2026	Only updates to existing records (and Admin rollbacks) can be scheduled; creating or deleting a record is never scheduled.
05-Nov-2026	Addresses can be IPv4 or IPv6: the portal creates A records for IPv4 and AAAA records for IPv6.
05-Nov-2026	Notifications are in the portal only; no e-mail.
05-Nov-2026	Proposal v2.2 is the agreed baseline for all updates since v2.0. Its points need no separate Bank confirmation here: one approval, direct Checker/Admin changes, versioning and rollback, checking directory users at every sign-in.
05-Nov-2026	Technology: Java/Spring Boot and React, as in proposal v2.2.

15. Open points

All open points from this document, grouped by who can close them. The proposed approach is the recommendation unless decided otherwise.

15.1 For the Bank (SBI, via F5)

#	Open point	Why it matters	Proposed approach
B1	MFA deviation (RFP J1.2 asks MFA for all users).	Compliance gap.	No MFA, as decided on 03-Nov-2026.
B2	Bank IS password policy for local accounts; lockout values; session rules (idle timeout, maximum length, simultaneous sign-ins).	Security policy compliance.	12+ characters, mixed; 90 days; 5 tries / 15 min; 30 min idle.
B3	LDAP details: LDAPS URL, base DN, user filter, service account, attributes, test users. Is the LDAP user ID the employee ID?	Needed to connect the real directory.	uid = employee ID.
B4	ITSM: product, API and depth; must an ITSM approval also gate execution?	Change governance.	Ticket and status only; portal approval gates.
B5	DNS form fields and rules; further record types needed (CNAME, MX, TXT …); object scope (records only, or Wide IPs, pools, monitors).	Scope and effort.	A, AAAA and PTR; others on request.
B6	Volumes (zones, records) and sample DNS data.	Sizing and performance.	—
B7	SIEM product, log format and transport; log retention periods.	SIEM integration; compliance.	JSON lines; 365 / 180 days; DB 7 years.
B8	RTO 1 min: one server, or the whole DC? TLS 1.3 deviation for the F5 management API (TLS 1.2).	Architecture and compliance.	Two RTO values; TLS 1.2 to F5.

#	Open point	Why it matters	Proposed approach
B9	Reports needed beyond the dashboard (J1.11); records per department or per partition (RPT-03).	RFP J1.11.	Records per partition; reports on request.
B10	Branding: F5 logo or SBI branding.	UI.	F5 logo for now.

15.2 For F5

#	Open point	Why it matters	Proposed approach
F1	REST API documentation (items listed in the request already sent): records incl. PTR, sync groups, transactions, save-config, auth, limits, error format, versions.	Needed for the F5 integration.	—
F2	BIG-IQ Centralized Management or direct BIG-IP (iControl REST)?	How the portal connects.	Whichever F5 confirms.
F3	How F5 partitions apply to the DNS objects the portal manages.	Partition model (PAR-01).	One partition per record.
F4	Reading live configuration: drift check before rollback; full sync and reconciliation (J1.5).	VER-05, DNS-08, PAR-05.	—
F5	UAT access (BIG-IP VE) on the deployed version; least-privilege API account.	Testing and security.	—

15.3 Design points to confirm

#	Open point	Why it matters	Proposed approach
D1	Can the Checker who approved a scheduled update cancel it, or only the maker (SCH-08)?	Workflow behaviour.	Only the maker.
D2	Should a scheduled update lock its record (SCH-06)? If its time passes before approval: refuse, or run at once on approval (SCH-03)?	Workflow behaviour.	No lock; refuse late approval.
D3	Suggest partitions from the department when the Admin gives access (ONB-08).	Faster, consistent access.	Include if wanted.
D4	Further session controls: maximum length, simultaneous sign-ins, Admin view of active sessions, ending sessions on password change (AUTH-09).	Security hardening.	Decide together with B2.